AI Music Generator (CodeAlpha Internship - Task 3)

An end-to-end deep learning project that ingests MIDI musical data, preprocesses notes using music21, trains an LSTM neural network model to learn sequential patterns, and generates original musical compositions.

Features

- **Data Preprocessing:** Extracts pitches and chords from standard MIDI files.
- **Deep Learning Architecture:** Built using stacked LSTM networks and Dropout layers via TensorFlow/Keras to prevent overfitting.
- **Interactive Web Interface:** Built using Streamlit for live demonstration, model training, and instant audio rendering/downloads.

Installation & Setup

1. Clone the repository:
   git clone https://github.com/your-username/AI_Music_Generator.git
   cd AI_Music_Generator

2. Install dependencies:
   pip install -r requirements.txt

3. Run the Web Interface:
   streamlit run app.py